

The Breakfast Paradox: Leverage Points, Intervention Design and Recommendation

Team Invictus

This report covers Phases 3 and 4 of the Breakfast Paradox project: why IIIT Hyderabad students get billed for breakfasts most of them never eat, and what could change that. Activity 9 works out where and how the system can be changed, producing the Leverage-Point / System Intervention Map and the Design Principles. Activity 10 takes it further. It designs and tests interventions against the system as a whole rather than any one stakeholder's interest, and delivers the intervention concepts and prototypes, the scenario model and evaluation framework, and the recommended intervention.

We start from the Phase 2 reframe. The Breakfast Paradox isn't a broken system; it's a stable equilibrium, and three structures hold it there. Registration is an automatic default with a narrow way out: five cancellations per meal type per month, each made at least four days ahead, with no way to undo the charge beyond that (confirmed). Billing fires at the moment of booking, not at the moment of eating (confirmed). And the gap between the two is measured but never routed anywhere. The CDS scan record and the operators' own books both capture it, but neither reaches whoever sets the registration, billing or menu rules (confirmed).

Underneath the kitchen side sits one intermediate mechanism: turning registrations into a cooked first batch. Kadamba cooks a fixed 70 percent of registrations per line (single-sourced). Bakul (Vijayalakshmi Caterers) cooks 70 to 80 percent by item, a figure its manager sets each week from his read of past attendance, never letting it drop below about 70 percent even on the quietest weekends (single-sourced). Yuktāhār plans from its own books. Every kitchen can top up a batch, but none will shrink one. Whatever the gap produces gets absorbed long before it becomes a signal anyone notices: through resale, top-ups, reuse, the meals of outside diners and staff, late service, holding and reheating, and finally the bin. Each actor is responding rationally to the incentives and information it has; the outcome nobody wants is what happens when those responses connect. The design opportunity, then, is to give the system a way to notice its own cost, by routing the records that already exist to the people who set the first batch and the rules, and by testing the beliefs that keep the working goal fixed on the registration count.

Claims here carry the project's confidence vocabulary. *Confirmed* means read directly off a record (the Kadamba April 2026 breakfast scan export, the photographed Yuktāhār registers, posted rates) or seen on site. *Single-sourced* means it rests on one account: the Kadamba operator's figures are first-hand, so the numbers themselves aren't in doubt, but they remain his account of his own practice; the Bakul manager's figures come from a team member's synthesis of an interview with him. There are no Bakul records of any kind, so no Bakul effect is sized anywhere in this report. *Candidate* means an inference built by combining sources, and *assumed* means a judgement offered without direct evidence. Three more labels apply only to Activity 10. *Projected* marks every scenario-model output: arithmetic run on real records under stated assumptions, not an observed result. *Simulated* marks every stakeholder

response: a reasoned expectation, not feedback from the actor itself. *Planned* marks every co-design test, none of which has run yet. Loop and chain IDs (L1 to L17 and L3b) and their verdicts carry over from the Phase 2 causal loop analysis. Leverage depth is written as a position on a twelve-point scale, counted from the deepest: place 2 is mindset, place 3 the goals of the system, place 5 its rules, place 6 its information flows, and place 12 its parameters.

1 Activity 9: Define Design Principles and Intervention Opportunities

Activity 9 works out where and how the breakfast mess system could change. It produces two deliverables, the Leverage-Point / System Intervention Map and the Design Principles, and moves through the activity’s four methods in order: identifying leverage points and reading them against archetypes and rank; exploring changes to rules, roles, incentives, information flows and processes; setting the design objective against the case for the status quo; and locating fourteen intervention points. The map and the seven principles close out the section.

1.1 Identify Leverage Points

Each leverage point sits on the twelve-point leverage scale, counted from the deepest, so place 1 is the power to transcend paradigms and place 12 is parameters. Place 5 (rules) is used only where a rule itself changes: the default and its exit, who decides, what a booking costs, or the standing rule that keeps another lever in place. Place 9 (delays) is reserved for flattening the closing crest, which no lever here attempts. Places 11 and 12 only appear as settings inside a deeper lever. Lowering Kadamba’s 70 percent ratio by decree would simply move the cost from surplus into run-outs at the crest, and the cancellation cap, raised from five to ten during a gas shortage and now back at five (single-sourced), only changes how many exits the default allows.

A lever is ruled out if it works by removing a mechanism that protects someone: resale on Mess Cell, reuse of leftovers, the meals of 30 to 50 outside diners and about half of Kadamba’s own staff, the walk-in buffer, late and back-door service, outside venues, informal accommodation. The over-cooked first batch, repeated holding and reheating, and the bin protect no one, so they are fair targets. Feasibility is judged for a student team, which can propose, pilot and evidence a change but not enact a rule. High means a prototype or pilot with an actor already engaged; medium, an evidenced proposal to an existing forum; low, a proposal only, because either the holder can’t be reached or the lever depends on a fact nobody yet knows.

Table 1. The twelve leverage points

Leverage point	Place	Holder of the power	Feasibility
LP1 Build the missing routes	Place 6 of 12 (information flows)	CDS (scans), operators (books); receiver unnamed	High to prototype, medium to deliver
LP2 Change the breakfast default and its exit	Place 5 of 12 (rules)	Registration rule owner (Mess Office or CDS, unconfirmed)	Medium

Leverage point	Place	Holder of the power	Feasibility
LP3 Operator goal: surplus is a cost	Place 3 of 12 (goals) for purpose; place 8 of 12 (balancing loops) for the target	Prism management; Bakul manager and firm	Low to medium
LP4 A standing operator forum that runs its own trials	Place 4 of 12 (self-organisation)	Operators; CDS as convenor	Medium
LP5 Record-calibrated first batch with a daily learning record	Place 6 and place 8 of 12; allowances at 11 and 12	The operator	High (Yuktāhār), medium (Kadamba), unsizeable (Bakul)
LP6 Crest-aware top-up and a staged late batch	Place 6 of 12 (card) and place 10 of 12 (stocks and flows)	Operators; Mess Office or CDS for the window	High (Kadamba), low (Bakul)
LP7 A cost trigger for attention, carried on the oversight line	Place 8 of 12 (balancing loops); rule at 5	Escalation pathway; facility team	Medium, after LP1
LP8 Operator voice in the menu	Place 6 of 12 (information flows); slot at 5	Menu committee (unconfirmed)	Medium
LP9 A named owner of booked, cooked and eaten (held)	Place 5 of 12 (rules: who decides)	CFS and CDS	Low
LP10 What a booking costs: billing and payment basis (held)	Place 5 of 12 (rules: incentives)	Billing owner; contract holder	Low
LP11 Surface and test the governing mental models	Place 2 of 12 (mindset or paradigm)	Each belief's holder	Medium
LP12 Change the goal the system actually seeks	Place 3 of 12 (goals of the system)	CFS and CDS	Low to medium

LP1: Build the missing routes. Kadamba recorded 32,162 breakfast registrations in April 2026, of which 37.1 percent were eaten, and Yuktāhār's September breakfast ran at 17.7 to 40.2 percent against lunch at 56 to 89 percent (confirmed). No route from any record to any rule owner has been found, a confirmed absence, and Yuktāhār's two books even disagree on 3 September: 77 eaten against 110 (confirmed). The fix is a standing breakfast view per hall, weekday and line (vegetarian and non-vegetarian) of booked, cooked, eaten and left over, call it the ledger, sent to the rule owner and back to the operators, built on one agreed definition of "ate". It also carries a weekly note of exams, events and holidays into the day-before planning meeting, since today these reach the Yuktāhār kitchen only by word of mouth (confirmed). The ledger has to accept paper, because Yuktāhār already dropped a commercial ERP system once (single-sourced), and it has to be blame-free. Delivery waits on a receiver, because two different people are on record as head of CDS. A monthly bill line showing breakfasts paid for and not eaten is a student-facing variant, conditional on LP2: a signal with no available response would only confirm that the charge can't be avoided.

Figure 1. The records already exist on paper. (a) Yuktāhār's attendance register for September 2026, breakfast on the left page and lunch on the right, each row closing with a turnout percentage. (b) Its Wastage Book for the Thursday of weeks one and three (3 September

The left photograph shows a student's handwritten food diary for Thursday, 23/9/16. The diary is organized into columns for different food categories and rows for specific items. A student's finger is pointing to the 'Green chutney' row. The right photograph shows another student's handwritten food diary for the same date, 23/9/16. This diary includes a summary of intake at the bottom, stating 'Water intake = 0' and 'Coffee intake = 2.4'.

LP3: Operator goal: surplus is a cost. Purpose, moving from “cook for every booking” to “feed whoever comes, at least cost”, sits at place 3; the numeric target inside L11 sits at place 8. Operators state different purposes under the same rules. Yuktāhār names quality, less waste, lower cost, standardisation and profit; Kadamba cooks on registrations “because it is a

responsibility and they are getting paid for it”; the Bakul manager’s rule is “there should not be a shortage of food, even if some wastage occurs” (all single-sourced). At Bakul, a run-out routes a complaint straight up to the managing director while surplus stays silent, so the rule is really the kitchen-level form of a system that only reacts to what’s loud (candidate). The fix is a wastage board, a target and a purpose stated by the operations manager, with the target anchored to the hall’s best recorded performance, Yuktāhār’s rate, or the institution’s own goal, never last month’s average, which just drifts down. At Bakul it can only hold as a cut to surplus with no rise in run-outs. If operators are paid on registered plates, cutting surplus lowers cost while revenue holds, though the rate may get renegotiated once savings show, and the cheapest cut is smaller portions, so the target always has to be paired with grams per plate, taste and run-outs (P7). The team can put a cost-per-line case to Prism’s operations manager, who hasn’t yet been interviewed.

LP4: A standing operator forum that runs its own trials. The three operators already run three different planning logics: Yuktāhār a matching-week lookup and weekday register, Kadamba a fixed 70 percent, Bakul 70 to 80 percent by item set weekly from past attendance and never below about 70 percent, topped up by phoning the Hafizpet kitchen (single-sourced; Yuktāhār’s books confirmed). Yuktāhār’s books are photographed (confirmed), and its waste fell between August and December once weighing was made blame-free (single-sourced, no before-and-after figures). Kadamba keeps gram standards but no learning record, and the Bakul operator believes Kadamba fills 90 to 95 percent when the records actually say 37 percent at breakfast (confirmed). The fix is a forum convened by CDS and hosted by Yuktāhār’s two project managers, where each hall runs its own trial and the ledger shows the results. Simply exchanging books would sit at place 6; an empowered forum with the ledger as the selection mechanism is place 4. What has to transfer is the preconditions: blame-free weighing, daily discussion, and a purpose that treats waste as cost. Kadamba’s layered chain of command carries a reprimand risk that Yuktāhār’s flat team doesn’t have.

LP5: Record-calibrated first batch with a daily learning record. L11 closes at Yuktāhār, runs as a candidate instance bounded by a floor at Bakul, and is simply absent at Kadamba. Creating it at Kadamba sits at place 6; strengthening it where it already runs sits at place 8. Kadamba cooks 70 percent of registrations per line, but biryani lunch runs at 100 percent (single-sourced), so the ratio is following expectation, not a record. April turnout came in at 32.4 percent veg and 48.5 percent non-veg (confirmed), meaning the batch is roughly 2.2 times what was eaten on the veg line and about 1.4 times on non-veg (candidate arithmetic). The fix sets the batch as the gram standard multiplied by expected turnout per hall, weekday and line, plus a named allowance for outside diners and staff and a float for bonda, puri, vada and idli, starting on the veg line, with a one-page daily paper record read the next day. Kadamba already cooks about 30 extra portions for outside diners, and about half its staff eat there too (single-sourced), so their meals and the raw reserve need to be budgeted in. The lever can’t yet be sized precisely: a 70 percent batch against about 398 eaten implies roughly 350 surplus portions (candidate), which sits oddly against Kadamba’s own stated 5 to 10 percent waste figure. At Bakul, which cooks 70 to 80 percent by item and keeps food for about 70 percent even on very low weekends against a turnout the manager puts at most around 60 percent (single-sourced), item-level cooked-and-eaten figures would let us check his weekly adjustment; his floor stays a named constraint regardless. It is a quick fix, paired with LP1 against the L17 risk and with LP6 against run-outs.

LP6: Crest-aware top-up and a staged late batch. At Kadamba, 26.5 percent of April scans land between 9:15 and 9:30, rising to about nine a minute by 9:29, and 9.2 percent, about 37 a day, arrive at or after 9:30, the latest at 10:07 (confirmed). The operator's own estimate of about 150 diners in the last ten minutes (single-sourced) overstates this, and is a candidate driver of why the hedge exists. Yuktāhār projects from the first half hour (single-sourced), when only 15.3 percent of Kadamba's scans have even arrived (confirmed). The fix has three parts: a weekday arrival card telling the supervisor at 8:00 what share of the day has already shown up (place 6); a late batch started shortly after 9:00 based on it (place 10, since it changes when stock gets created); and a grace period after 9:30, always paired with the late batch. Place 9 isn't claimed here. The share seen by 8:00 runs from 13.5 percent on Sundays up to 22.5 on Wednesdays, and the share after 9:15 runs from about 32 percent (Tuesdays, Wednesdays) to about 46 (Sundays) (confirmed; used as factors, candidate), so a single flat factor would misread Sunday and Wednesday by about a quarter. None of this transfers to Bakul. The manager places his rush at 8:00 to 8:30 and aims to have food ready for about 50 diners at 9:20 (single-sourced), and a top-up there needs a phone call, cooking at the Hafizpet kitchen and a drive of about 8 km, roughly 20 minutes if called early and 40 otherwise, with the end-to-end lead time never actually measured. The Bakul version of this lever is really about the size of that held reserve and the latest call that can still land, set from measured data. The reserve is a hedge, not an absorber (candidate). LP6 is part of the quick fix.

LP7: A cost trigger for attention, carried on the oversight line. Loud menu fatigue got fixed through the Student Council, Parliament and CDS Committee (confirmed, L4); the much larger breakfast gap never entered that pathway at all. The facility team tastes every Kadamba meal alongside two Prism staff (confirmed, observed), but never checks quantity. The fix adds a standing threshold, say, cooked-against-eaten beyond a set level for several weeks running, that sends the case to the Committee alongside the existing complaint trigger. It is anchored to the stated goal, lunch turnout or Yuktāhār's rate, never the hall's own past, and it carries run-outs alongside surplus. The cheapest way to carry it is one extra line item at the tasting. Because an inspector writing down quantity can look like an audit, a breach triggers a review, never a deduction, and the line adds no delay to safety.

LP8: Operator voice in the menu. At Kadamba, uttapam, set dosa and upma waste the most while puri, bonda, vada and idli move well (single-sourced), yet it is students or the committee who choose the menu, not the kitchen. The Bakul manager wants menu input and already knows what sells, like pav bhaji, and which pairings he thinks work, like kachori with chole (single-sourced); that students would eat more of familiar pairings is his view, untested with residents. The fix is a standing operator slot at each two-week menu cycle, bringing item waste and run-outs to the table, which builds the missing route L16 as the quiet counterpart to L4. Input gets read alongside consumption records and checked against residents' actual views; it is a voice, never authority, and no pairing view enters the record as student evidence on its own.

LP9: A named owner of booked, cooked and eaten (held). Decision rights are split at least five ways, across registration, billing, menu, cooked quantity and the academic calendar (confirmed). The actor who sets registration and billing rules bears none of the cost those rules produce, so a named owner, not a committee, would put the consequence where the decision actually gets made, and would inherit LP1. Yuktāhār already has a lighter version of this in its Wastage Register keeper. The lever stays held until the head of CDS is identified.

LP10: What a booking costs: billing and payment basis (held). Attendance-tracked billing ran during an LPG shortage and some holiday and fest periods, and it worked (confirmed). Operators are paid monthly as a lump sum on plates, with deductions (single-sourced, noisy audio). The Bakul manager says plates served (single-sourced), where “served” is the normalised wording of one remark; Kadamba’s “paid for it” is a hint, not a contract term. No contract has been seen, and the basis may differ by operator. The lever stays held because a billing change made without knowing the payment terms could push the operator incentive the wrong way, and student reliance on resale income needs checking first.

LP11: Surface and test the governing mental models. The beliefs holding the default, the fixed ratio and the unrouted gap in place sit at the base of the Iceberg chains, and nobody can change them on someone else’s behalf. Eight of them run into a record that doesn’t fit; the four Bakul beliefs meet no record at all, so the table below names the record that would actually test them.

Table 2. Governing beliefs and the records they meet

Belief (holder, confidence)	What the record shows	Sign of a shift
Registration count stands in for demand (administration; candidate)	37.1 percent eaten at Kadamba in April; flat count against 27.3 to 42.1 percent by weekday (confirmed)	Rule owner cites booked against eaten
“Attendance billing is for emergencies” (candidate)	It ran and worked in the shortage and some fest periods (confirmed)	Discussed for Sunday breakfast
“Cooking to registrations is what we are paid for” (Kadamba; single-sourced)	Yuktāhār plans the same booking from records (confirmed)	Batch set in expected eaters
“Breakfast is about 50 percent” (Yuktāhār site head; confirmed)	His register shows about 28 percent (confirmed)	Stated and planned figures match the register
“Student fluctuation is very high” (Yuktāhār; confirmed)	Weekday arrival profile is regular (confirmed; projection candidate)	A staged cook is tried and tracked
About 150 in the last ten minutes (Kadamba; single-sourced)	About 79 scans a day from 9:20 to 9:29 (confirmed)	Supervisor’s prior guess comes close
“We don’t have any opportunity to nullify it” (students; confirmed)	The exit exists but is narrow	Skip use rises after LP2
“They are catching us” (Yuktāhār staff; single-sourced)	Dissolved once weighing was made blame-free	Sheets filled unprompted
At most about 60 percent come (Bakul manager; single-sourced)	No record; other halls show about 37 and 28 (confirmed). Test: Bakul scans	Batch follows Bakul’s scans
Students skip for taste, menu or sleep (Bakul manager; single-sourced)	No resident asked; six students elsewhere cite late sleep most (confirmed). Test: residents’ reasons	Menu changes follow residents’ reasons
Pairings, not dishes, are the problem (Bakul manager; single-sourced)	No item or pairing record. Test: cooked and eaten by pairing	Proposal arrives with figures
“There should not be a shortage of food, even if some wastage occurs” (Bakul manager; single-sourced)	No record of how often a smaller batch would run short. Test: daily run-out record	Floor and reserve set from that record

Every session starts from the ledger, so the question on the table is what the record shows, not whose view wins. Sessions pair advocacy with inquiry and point at specific anomalies, such as

50 against 28, though at Bakul the record itself has to come first, since there isn't one yet. Effort goes to the actors already open to the data, with the holders of tested views made visible.

LP12: Change the goal the system actually seeks. The CDS poster states the institution's goal as minimising food wastage through advance meal planning (confirmed). But the goal the system actually seeks is the registration count: students are billed on it (confirmed), Kadamba cooks on it (single-sourced), and reading these together as the real working goal is a candidate reading. The measure of the stated goal is already being recorded and read by no one (confirmed). The fix makes booked-against-eaten and cooked-against-eaten, per hall and line, the real indicators, read alongside quality and run-outs, since a wastage goal read alone could just as easily be met by running short (candidate). CFS and CDS carry this by restating their own goal and asking for its indicators. LP12 is the move from the registration count to cooked-against-eaten; the other levers are what make that move possible.

Identify System Archetypes

Of six structures checked, only the first three read as actually present.

Table 3. Archetype reading

Archetype	Loops and evidence	Confidence	Way out and levers
Shifting the burden	Quick fix L11, L12, L9, L1 (resale); fundamental fix withered (L10 route, L13 upward link, L2, reverted L6); side effect L17. Resale replaces Skip Meal; attendance billing reverted (confirmed)	Candidate; the link from kitchen competence to less pressure upstream is assumed	Strengthen the fundamental fix (LP1, LP2, LP12, LP10); name the quick fix (LP5, LP6); P5
Balancing process with delay	L12, L14. Top-up of 10 to 15 minutes against the crest; at Bakul a 20 to 40 minute drive and a held reserve	L12 confirmed; delays single-sourced; overshoot candidate	Be more responsive; do not over-correct (LP6)
Fixes that fail	December 2024 cap tightening; a candidate ratio ratchet on L14 (premise only); reheating branch of L13. No backfire observed	Candidate	Make the implicit target explicit (LP3, LP5, P2)
Eroding goals	Stated goal against the count; "this cannot be helped"; 50 against 28. No lowered target on record	Not established; a risk (assumed)	Anchor standards outside (LP3, LP7, LP12)
Accidental adversaries	Operators and rule owner (L10, L16). No antagonism found	A risk if the ledger reads as audit	Map the structure together (LP4, LP11, P2)
Commons or success to the successful	L3b (candidate); Kadamba full, Bakul under-registered; no Bakul or Palash records	Low, outside scope	None; LP2's hall-slot condition guards L3b

The first archetype orders the whole set. LP5 and LP6 are the quick fix, worth doing because students eat the food that gets saved and it buys time, but it needs to be named as such in the ledger, since a better-run kitchen presented as the answer would stop anyone looking for the fundamental fix (the L17 risk). LP1, LP2, LP11 and LP12, with LP10 once released, make up the fundamental fix. The system is also chasing the wrong goal, the sharpest finding of the

whole problem analysis, and that's what LP12 answers. One past change already pushed the wrong way: the cap was tightened in December 2024 to ease Mess Office workload, narrowing the exit (single-sourced; effect on no-shows candidate). An earlier reading, that Bakul raised its ratio from 70 to 80 percent after run-outs, is withdrawn; the figures simply describe variation by item and day.

Ranking: Depth and Feasibility

Depth combines place with role, so a fundamental-fix lever outranks a quick fix sitting at the same place; feasibility is judged for a student team in the weeks remaining.

Table 4. Ranking of leverage points

Rank	Lever (place)	Role	Depth / feasibility	Gated on
1	LP1 (6)	Fundamental, enabling	High / high to prototype	A named receiver
2	LP11 (2)	Fundamental	Very high / medium	The ledger
3	LP2 (5)	Fundamental	High / medium	Lowering the kitchen's count
4	LP12 (3)	Fundamental	Very high / low to medium	CFS and CDS
5	LP3 with LP4 (3, 8; 4)	Links both	High / low to medium	Access to Prism management
6	LP5 with LP6 (6, 8; 6, 10)	Quick fix	Medium / high to low by hall	Waste base; Bakul records
7	LP7 (8)	Fundamental	Medium / medium	Facility team's parent body
8	LP8 (6)	Fundamental	Medium / medium	Records and resident views
Held	LP9 (5)	Fundamental	High / low	Head of CDS
Held	LP10 (5)	Fundamental	High / low	The plate basis

LP1 ranks first even though it's shallower than LP11, LP12 and LP2, because every other lever needs its input to work. LP11 is the deepest lever but depends on the ledger existing first. LP2 is the deepest rule lever that isn't gated on the plate basis. LP12 comes fourth because nobody who could carry it has engaged yet, and a goal without indicators or an exit doesn't do anything on its own. LP3 and LP4 are what carry the goal into the operator's own routine: Kadamba isn't missing information, it's missing a reason and a routine. LP5 and LP6 are the fastest physical levers, but they are the quick fix, unsizeable until the waste conflict is settled, and at Kadamba they only land through LP3 and LP4. LP7 outranks LP8 because it gives governance an actual response to cost. LP9 and LP10 are held, not dropped.

1.2 Explore Changes to Rules, Roles, Incentives, Information Flows and Processes

Each change starts from the current state and implements a leverage point.

Table 5. Changes by type

Type	Current state	Candidate change	Lever
Rules	Automatic registration; five cancellations per meal type per month, four days ahead (confirmed)	Shorter lead; evening skip that lowers the count; Sunday opt-in; no hall-slot penalty	LP2
Rules	Attention follows loud complaints (L4); billing at booking; back-door entry after 9:30 cannot be refused (single-sourced); no protection for recorders	Standing cost threshold; consumption billing as already run; grace period with late batch; no deductions from recorded figures	LP7, LP10, LP6, LP1
Roles	Facility team checks safety and taste; nobody convenes operators or owns the outcome; menu chosen without operator input	Quantity line at tasting; CDS-convened forum; named owner; operator menu slot; CFS and CDS as goal carrier	LP7, LP4, LP9, LP8, LP12
Incentives	A non-eater pays anyway; Skip Meal returns nothing (confirmed); recording read as being caught; shortage loud, surplus silent (candidate, L14)	A later exit that removes the charge; surplus shown as cost; blame-free recording; late batch or sized reserve	LP2, LP3, LP5, LP6
Information flows	Records reach no rule owner; two definitions of “ate”; calendar by word of mouth; no kitchen-to-menu route (L16)	The ledger; bill line after LP2; weekly calendar note; item waste into menu cycle	LP1, LP8
Processes	Fixed 70 percent at Kadamba, weekly judgement with floor at Bakul, no record; whole hedge cooked at start	Calibrated batch with daily record; arrival card; late batch after 9:00; safety steps unchanged	LP5, LP6

The widest rule change costs the kitchen the least, since the four-day lead only protects dry goods anyway. The key incentive finding here is conditional. Where payment is on registered plates, cutting surplus lowers cost while revenue holds. That route is open right now, though it closes if the rate gets renegotiated and turns harmful if it’s achieved through smaller portions. Where payment is on served plates, the gain is bigger, but so is the pressure to serve more, and if the Bakul manager’s account holds, Bakul is already absorbing the cost of uneaten portions and accepting it to avoid shortage (candidate). P7’s paired measures guard against both cases.

The Natural Comparison

Kadamba and Yuktāhār share registration, billing and, as far as is known, tendered plate payment. Yuktāhār plans from records, re-cooks by run-out time and weighs blame-free on paper (books confirmed, practice single-sourced); Kadamba cooks a fixed ratio instead. So a learning practice is clearly possible inside the current rules, but the comparison can’t prove the cause, because the two halls differ in five other ways.

Table 6. Rival explanations

Rival explanation	Kadamba	Yuktāhār
Scale	About 1,070 registrations a day	About 230 to 400 a day
Menu and lines	Veg and non-veg	Veg only, with Jain service
Firm and layers	Prism, four-level chain	ABC Hospitality Services, flat team
Month of records	April 2026	September 2026
Measurement	CDS scan export	Handwritten registers

So two claims stay candidate: that the real difference is goal and culture, and that LP3 through LP6 need no rule change to work. The gap is also about equally large at both halls, 63 percent unused at Kadamba against about 72 percent at Yuktāhār (confirmed). Kitchen practice reduces waste and run-outs, but it doesn't move the booking; only LP2, LP10 and the goal-and-belief levers do that. Bakul, a third regime with its own rival explanations (an off-site kitchen, a different payment report), can't enter this comparison without records of its own.

1.3 Develop Design Principles

The objective is a breakfast system where registration matches each student's actual intent, the kitchen cooks to expected eaters, and the cost of any gap reaches the people who set the rules. It doesn't mean eliminating registration, resale or all waste. Registration gives the planning count, resale recovers money, and some surplus is simply the price of never running out. The aim is to make over-booking and over-cooking exceptions someone notices, not the norm. The guiding idea follows straight from the archetype reading: change the architecture, don't compete at the kitchen layer.

The status quo persists because it pays each actor something.

Table 7. The case for the status quo

Actor	What it gives	Cost of change	Cost of not changing
Students	No decision; a guaranteed seat; resale income	A nightly decision; less resale	Paying for uneaten breakfasts (63 percent of Kadamba's April registrations); reheated food
Prism (Kadamba)	Revenue possibly on registered plates; few run-outs	Possible revenue fall; a daily record	Food cooked that no one eats; reheating
Bakul operator	No shortage, secured by a floor and reserve (single-sourced)	A check on the manager's judgement	Food held hot for hours, softening items such as aloo paratha
CDS	Low workload	A standing ledger; portal changes	A stated goal it cannot show it meets
Staff and outside diners	Food from surplus	Meals must be budgeted	Meals stay a by-product
CFS	A predictable budget	A budget varying with consumption	Spending on food nobody eats

The levers keep what matters to each actor, the seat, the resale, the absorbers, run-out protection, while changing what actually drives over-booking. CDS and the Mess Office, who

gain the most from low workload, own the rules the fundamental fix touches, so LP1, LP11 and LP12 need to reach them before any rule proposal does. The principles themselves make up Deliverable 2, below.

1.4 Identify Potential Intervention Points

The fourteen intervention points follow a breakfast from booking through to the bin, then the governance sitting around it.

Table 8. Intervention points

IP	Where and when	What happens today	Levers
IP1	Default registration, each cycle	Every student registered, on campus or not	LP2, LP11
IP2	Cancellation exit, four days ahead	Five cancellations; Skip Meal removes no charge	LP2
IP3	Kitchen's count, evening before	Registrations; Skip Meal effect in conflict	LP2
IP4	Day-before plan; Bakul's weekly review	Lookup, fixed ratio or judgement with floor	LP5, LP1, LP3, LP4, LP11
IP5	First batch and safety, 5:30 to 7:30; Bakul from about 3 AM	Whole hedge cooked before any signal	LP5, LP7
IP6	Tasting and weekly inspection	No quantity check	LP7
IP7	First half hour, 7:30 to 8:00	Flat projection or judgement	LP6
IP8	Counter check, about 9:00; Bakul phone top-up	10 to 15 minutes; 20 to 40 at Bakul	LP6
IP9	Crest and back door, 9:15 onward	A quarter of scans in fifteen minutes	LP6
IP10	Surplus after close	Held, reheated, reused; eaten by outside diners and staff; some returned to the Hafizpet kitchen	LP5
IP11	Daily record	Wastage Book at Yuktāhār only	LP5, LP4
IP12	Disposal	Bin to contractor at Kadamba	LP1
IP13	Records to rule owners	Export reviewed by no one	LP1, LP7, LP9, LP11, LP12
IP14	Menu cycle, two-weekly	No operator input or forum	LP8, LP3, LP4, LP11

Monthly billing and vendor payment carry LP10 and LP1's bill line. Bakul needs no separate intervention point of its own, since its dispatch is IP5 and its phoned top-up is IP8 with a longer delay attached.

Sunday Breakfast: Where Three Levers Meet

Sunday has Kadamba's lowest turnout, 27.3 percent, against about 1,070 registrations (confirmed). A 70 percent veg batch runs about 3.1 times what Sunday diners actually ate, against about 1.9 times on Tuesdays (candidate arithmetic), and about 46 percent of Sunday scans arrive after 9:15 (confirmed). Yuktāhār's Sundays run at 17.9 to 24.1 percent (confirmed). The Bakul manager calls weekends "very low" (single-sourced) but doesn't attach

a figure, so this slice stays specific to Kadamba's veg line. Sunday is where LP2 (IP1 to IP3), LP5 (IP4) and LP6 (IP8, IP9) have their biggest effect, which makes it the natural first slice for Activity 10.

1.5 Deliverable 1: Leverage-Point / System Intervention Map

The map draws the breakfast system as one chain, from the automatic default through the day-before plan, the first batch, service and the surplus, down to the bin, with records and money running above it and governance at the top. Markers are coloured by place. LP5 and LP6 each appear once per part, LP3 is marked by its deeper part, and a dashed ring marks a held lever. Dashed red lines are routes that don't exist today and that a lever would need to build; the dashed green line is the learning link that runs at Yuktāhār and is absent at Kadamba. Grey tags carry the intervention points, with IP13 and IP14 sitting at the governance bodies the dashed routes would reach.

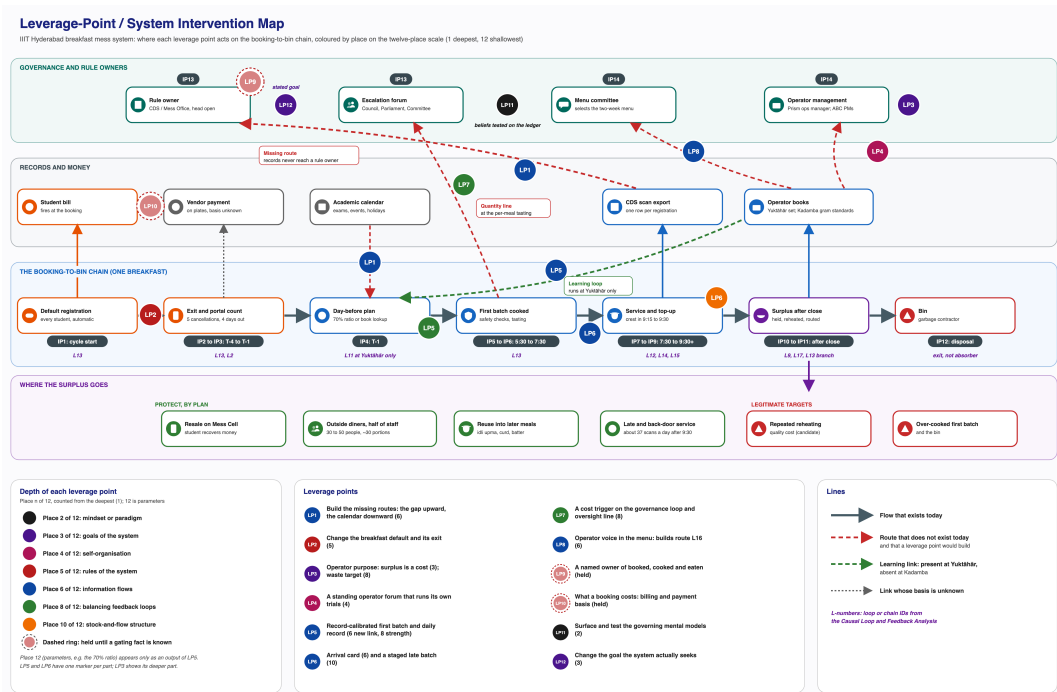


Figure 2. Leverage-point and intervention map for the breakfast mess system

Figure 2. Leverage-point and intervention map for the breakfast mess system

Every solid flow runs along the chain or up into a record, and stops there. Every lever that builds a route, sets a goal, tests a belief or convenes operators (LP1, LP3, LP4, LP7, LP8, LP11, LP12) sits on a route or body that doesn't yet do that work, and the rule levers (LP2, LP9, LP10) sit on the booking, the bill and the owner that the missing routes would reach. The only feedback loop actually working into the day-before plan is the learning link, confirmed at one hall out of three; Bakul's weekly review is a candidate second instance, bounded by its floor, which the drawing doesn't show yet.

Table 9. Map table

Intervention point	Levers (place)	Holder of the power	Absorber check
IP1 to IP3 Default, exit, kitchen count	LP2 (5); LP11 students' belief (2)	Rule owner and portal owner; student representatives	Resale need shrinks; none removed
IP4 Day-before plan	LP5 (6 Kadamba; 8 Yuktāhār, Bakul); LP1 calendar (6); LP3 (3, 8); LP4 (4); LP11 operator beliefs (2)	Operators; academic administration via CDS; Prism management	Budget outside diners, staff and reserve
IP5, IP10, IP11 Batch, surplus, record	LP5 (6, 8); LP4 (4)	Operator	Keep safety steps and raw reserve; diners fed by allocation
IP6 Tasting and inspection	LP7 (8; 5 for the rule)	Facility team; escalation pathway	Adds no delay to safety
IP7 to IP9 Projection, top-up, crest	LP6 (6, 10)	Supervisor; Yuktāhār team; Bakul manager; Mess Office or CDS for the window	Keeps back-door service, made explicit; Bakul reserve sized
IP12, IP13 Disposal, records upward	LP1 (6); LP7 (8); LP9 (5); LP11 (2); LP12 (3)	CDS, receiver unnamed; CFS and CDS	Bin is a target
IP14 Menu and management	LP8 (6); LP3 (3, 8); LP4 (4); LP11 (2)	Menu committee (unconfirmed); Prism management	Removes nothing
Monthly billing	LP10 held (5); LP1 bill line (6)	Billing owner; contract holder	Check reliance on resale

Across the whole set, only LP5 and LP6 actually touch an absorber that protects someone, and both are written specifically to keep it intact. LP5 budgets outside diners and half of Kadamba's staff by name and keeps reuse, the walk-in buffer and the top-up reserve; LP6 keeps them too and makes late and back-door service explicit with a late batch. LP2 and LP10 shrink resale by removing the need for it, not the market itself. Repeated reheating and the bin get recorded by LP1, reduced directly by LP3 through LP6, and reduced upstream by LP2, LP10, LP11 and LP12. No lever removes an absorber just to create a signal, but that's not the same as saying no lever costs anyone anything.

Table 10. Stakeholder impact check

Stakeholder	Gains	Bears	Levers
Students	A later exit; less reheated food; fewer late run-outs; an honest bill	A nightly decision; possible loss of resale income	LP1, LP2, LP5, LP6, LP10, LP11
Prism (Kadamba)	Lower food cost; fewer late shortages	Possible revenue loss and rate renegotiation; a daily record	LP2, LP3, LP5, LP6, LP10
Kadamba supervisor, staff, storekeeper	A clearer late-batch rule; issues matched to a plan	A 9:00 cook decision, a daily sheet, blame risk, a varying batch	LP5, LP6, LP7
Yuktāhār operator	A menu voice; a better breakfast anchor	Hosting time; its site head's figure set against his own register	LP4, LP5, LP8, LP11

Stakeholder	Gains	Bears	Levers
Outside diners and Kadamba staff	Meals by allocation, not accident	Being squeezed if the allowance lapses	LP5
Rule owner (CDS or Mess Office)	The gap visible in its own terms	A ledger, a threshold to answer, portal changes, an ownership role	LP1, LP2, LP7, LP9, LP12
CFS and facility team	Evidence the goal is met; a remit reaching quantity	A less predictable budget; being seen as auditors	LP7, LP10, LP12
Bakul operator	A check on its judgement; a menu voice; no revenue loss from LP2 if paid on plates served (candidate)	Requests for records it may not keep; any rise in shortage risk	LP2, LP3, LP5, LP6, LP8, LP11

Every actor here has a rational reason to prefer the present arrangement, so resistance is expected, not a fault to fix. The Kadamba operator stands to lose the most from LP2 and is the most needed for LP5, which is why LP3 and LP4 have to come before LP5 at Kadamba, and why the ledger has to reach CDS before any rule proposal does. At Bakul, the likeliest objection is to anything that raises shortage risk (candidate, held behind the unconfirmed plate basis).

1.6 Deliverable 2: Design Principles

Seven principles follow from the leverage points, the archetypes and the evidence. Each one is a test that any intervention concept in Activity 10 has to pass.

P1: Route before you measure, and route on paper. Connect the records that already exist to the decisions that need them, before adding any new instrument. The gap is already measured, in the CDS export, in Yuktāhār's books, in Kadamba's own stated waste share, and Yuktāhār already dropped a commercial ERP system over cost and staff readiness (single-sourced), so routes should start from a book page and travel on channels that already exist: the daily meeting, the tasting, the menu cycle. This rules out new apps, sensors or forecasting software as a first step, surveys that just re-measure what records already hold, and any design that needs operators to go digital.

P2: Make measurement blame-free, and judge it over months. No figure an intervention records should ever become grounds for a deduction against the staff who recorded it, and results should be judged over months, not days. Yuktāhār's weighing only worked once it was made blame-free and discussed daily, and the fall in waste took from August to December (single-sourced); one run-out after a smaller batch, judged over just a week, would be enough to bring the hedge straight back. This rules out quantity checks framed as audits, linking waste to deductions, and one-week trials.

P3: Change the default, the goal and the belief, not the people. Aim at the default on the student side, the goal on the operator and institution side, and the beliefs holding both in place, treating every actor as responding rationally to its own incentives and information. Students are registered by a default they never chose and hold the least power of anyone in the system; operators run the only balancing loops that actually work (L11, L12); the institution states one goal and pursues another. This rules out nudges and campaigns aimed at students, any logic of "students should just register more accurately," blaming operators, or changing the cancellation count while leaving the default untouched.

P4: Act on the first batch the day before, and on the morning only through the top-up.

First-batch levers act at the day-before plan; morning levers act only through smaller, later top-ups or, where one can't land in time, a sized reserve. None of this may cut waste by raising shortage risk, and scope stays per hall, weekday, line and, where records allow, item. At Kadamba the batch is fixed three and a half to four hours before the crest, and at Bakul cooking can start around 3 AM (single-sourced), while arrival curves and line turnouts (32.4 against 48.5 percent) differ substantially. This rules out same-morning signals aimed at the first batch, shortening safety steps, a single ratio for everything, cutting waste through run-outs, or moving one hall's rule onto another before it has its own records.

P5: Feed the absorbers by plan, name them as symptomatic outlets, and target only the over-cooked batch, repeated reheating and the bin. Any concept that shrinks surplus has to name who eats it and budget their meals, and the ledger should record absorbers as outlets of the gap, never as solutions to it. Thirty to fifty outside diners and about half of Kadamba's staff already eat from this same food (single-sourced), yet counting their meals as "not wasted" would just hide the gap again, the shifting-the-burden trap in miniature. This rules out restricting resale, closing the back door, cutting the batch without an allowance, or reporting absorbed surplus as a success.

P6: Measure success as cooked against eaten, run-outs and food quality, not turnout.

Success means the cooked-eaten gap, run-outs, food quality as actually eaten (holding time and texture, not just temperature and taste), and whether the cost reaches a rule owner at all; waste falling while run-outs rise is not success. Some skipping is entirely legitimate (confirmed across interviews), and Bakul's food, held at about 60 to 72°C, keeps its temperature even as aloo paratha softens (single-sourced). This rules out turnout targets, attendance drives, waste cuts sized before Kadamba's own 5 to 10 percent base is even known, and quality checks that read temperature alone.

P7: Measures are diagnostic and plural, never a single target. Every measure serves learning, not reward or penalty, and cooked-against-eaten always has to be paired with grams served per plate, taste and quality, and run-outs. A kitchen held to just one figure can meet it by cutting portions or letting the crest run short, and a vendor rewarded on it can meet it through the rate; paired measures close off both of those routes, and the targets in LP3 and LP7 exist to serve review, not to be gamed. This rules out a single headline target, any bonus, deduction or ranking tied to one ledger figure, and waste figures reported alone.

2 Activity 10: Design, Test and Evaluate Interventions

Activity 10 develops and assesses interventions that improve system-level outcomes rather than optimising for any one stakeholder's interest. Its three deliverables are the intervention concepts and prototypes (Deliverable 3, the first two sections below), the scenario model and evaluation framework (Deliverable 4, from the scenario model through the unintended consequences), and the final system design recommendation (Deliverable 5, after the refinement). The shifting-the-burden reading from Activity 9 sets the order throughout. The kitchen levers (LP5 and LP6) are the quick fix: they save food and buy time without touching why too many plates get booked and cooked in the first place. The route, the default and its exit, the goal, and the beliefs (LP1, LP2, LP12 and LP11) are the fundamental fix, so the recommendation runs them on fixed dates regardless of what the kitchen pilot shows.

2.1 Generate Alternative Interventions (Deliverable 3: Intervention Concepts)

These concepts target the three confirmed structures holding the equilibrium in place (registration as a standing default with a narrow exit, billing at the booking, and a gap that's measured but unrouted), plus the step where registrations turn into a first batch. Kadamba cooks a fixed 70 percent of registrations per line; Bakul (Vijayalakshmi Caterers) cooks 70 to 80 percent by item, set weekly by its manager's judgement and never below about 70 percent (both single-sourced); Yuktāhār plans from its own books. Every kitchen can top up a batch but none will shrink one. Concept G is a deliberate foil, carried through every test so that its failure gets shown rather than just asserted.

Table 11. The eight intervention concepts

Concept	Leverage points (place)	Intervention points	Role
A: The Breakfast Ledger	LP1, LP7; LP9 later (6, 8; 5 for the threshold)	IP13, IP6, IP12, IP4	Fundamental: route
B: The Kitchen Learning Kit	LP3, LP4, LP5 (3 to 8)	IP4, IP5, IP10, IP11, IP14	Quick fix, carried by operator levers
C: The Crest Plan	LP6 (6, 10)	IP7, IP8, IP9	Quick fix
D: The Sunday Exit	LP2 (5)	IP1, IP2, IP3	Fundamental: default and exit
E: The Operator Menu Slot	LP8 (6)	IP14	Fundamental: route
F: Billed at Consumption	LP10 (5)	Billing, outside the daily chain	Fundamental, held
G: Ratio Cut to 50 Percent	None (place 12 parameter)	IP4, IP5 as a number	Foil
H: The Shared Reading	LP11, LP12 (2, 3)	IP13, IP14, IP4, IP1	Fundamental: goal and belief

A: The Breakfast Ledger. Each week, the CDS scan export and the operators' books become one sheet per hall, by weekday and line, including the money charged for uneaten breakfasts. That charge line matters because it stops a falling surplus from being read as "problem solved": kitchen levers shrink surplus while the charge stays flat until a rule actually moves. A threshold on cooked-against-eaten, anchored outside the hall's own past, alerts the escalation forum alongside the existing complaint trigger, and a weekly note of exams and events flows down to planning. A routes L10, adds a second trigger to L4, counters L17, and brings L7's exam dips to the kitchen (P1, P2, P6, P7).

B: The Kitchen Learning Kit. Kadamba's fixed ratio becomes a day-before plan card built from expected eaters, with a named allowance for the 30 to 50 outside diners and about half of Kadamba's staff, plus a daily paper sheet read the next day. An operator forum, convened by CDS and hosted by Yuktāhār's project managers, carries over the preconditions behind Yuktāhār's Wastage Book: blame-free weighing, daily discussion, and surplus treated as cost. At Bakul, whose manager already adjusts weekly by item, the kit works more as a check than a new rule, with the floor kept in place until a record shows how often a smaller batch would

actually run short. Nothing there is sized yet. B creates L11 at Kadamba (a candidate second instance at Bakul), shortens L13's kitchen half, and reaches L8 through cost (P1 to P7).

C: The Crest Plan. At 9:00 the supervisor projects the day from a weekday arrival card and starts a late batch that lands by about 9:15, before the crest, so the first batch only needs to cover the earlier diners. A grace period after 9:30 covers late-batch food only. All of this rests on Kadamba's 10 to 15 minute on-site top-up. Bakul's top-up needs a phone call, cooking at the Hafizpet kitchen and a 20 to 40 minute drive (single-sourced), so its hedge is about 50 portions held at 9:20 (candidate), and its version of C is really about sizing that reserve and the latest call that can still land. C gives L12 sight of the crest, removes L14's reason to hedge, and bounds L15 (P4, P5, P6).

D: The Sunday Exit. Sunday, the lowest-turnout day at both recorded halls, is first drafted as opt-in: charged only if confirmed by 8 pm Saturday. A lighter variant keeps the default and adds a skip, open until the same hour, that removes the charge and lowers the kitchen's count. Other days keep the four-day lead and the cap of five cancellations, and no skip may ever cost a hall place. D acts on L13's booking end, on L2 (Skip Meal fails because a skip currently returns nothing) and on L1 (P3, P4, P5). Under opt-in, a student who forgets to confirm simply pays the walk-in rate.

E: The Operator Menu Slot. At each two-week menu cycle, operators bring the items that wasted most (uttapam, set dosa and upma at Kadamba, single-sourced) and those that ran out. The Bakul manager's view that pairings are the real problem (single-sourced) sets the guard here: operator input gets read alongside consumption data and residents' actual wishes, as a voice, never as authority. E supplies L16's missing link, which only becomes a loop once it's actually used (P1, P6).

F: Billed at Consumption (held). Breakfast gets charged on the scan, exactly as already done during an LPG shortage and some fest periods (confirmed), acting on L1, L2, L6 and L8 (P3, P5, subject to a check on resale income). It stays held because the operator payment basis is unconfirmed: the Bakul manager reports plates served (single-sourced), while the Kadamba operator says it cooks on registrations and is paid for that. The hold lifts only once each operator's basis is confirmed.

G: Ratio Cut to 50 Percent (foil). Kadamba's first batch drops from 70 to 50 percent by instruction, with no record, late batch or allowance attached. It changes no loop, and it breaks both P4 and P5.

H: The Shared Reading. Joint sessions around the ledger ask each belief holder what leads to their figure, and what would change it: the Yuktāhār site head's "about 50 percent" set against his own register at about 28, the Kadamba supervisor's crest prediction, the registration side's reasons for the automatic default. The Bakul manager's beliefs wait on a Bakul record. CFS and CDS are asked to carry the goal already on the CDS poster, minimising food wastage through advance planning, as two indicators, booked-against-eaten and cooked-against-eaten, read alongside P7's quality measures. H acts on the mental models beneath every chain and on the working goal behind L10, L13 and L17 (P2, P3, P6, P7).

2.2 Develop Scenarios and Prototypes (Deliverables 3 and 4: Prototypes and Scenario Model)

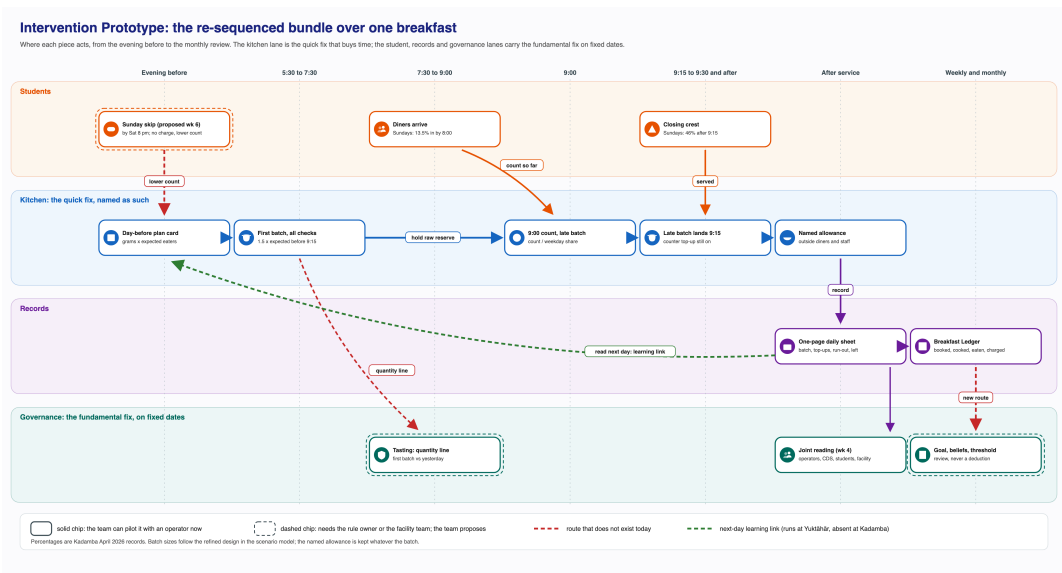


Figure 3. The re-sequenced bundle across one breakfast

Figure 3. The re-sequenced bundle across one breakfast: the kitchen lane is the quick fix, while the student, records and governance lanes carry the fundamental fix on fixed dates

Prototypes

Prototype A: the ledger sheet. One A4 page per hall per week, filled from the CDS export and the operator’s daily sheets. It can be written by hand and typed up later, just as Yuktāhār already does with its books (single-sourced).

Table 12. Prototype A, the ledger sheet, filled for Kadamba on Sunday 19 April 2026

Weekday	Line	Registered	First batch and top-ups	Eaten (scans)	Left over, and where it went	Run-out time	Charged, not eaten
Sunday	Veg	817	From the sheet	179	From the sheet: outside diners, staff, reuse, bin	From the sheet	$638 \times ₹48 = ₹30,624$
Sunday	Non-veg	252	From the sheet	94	As above	From the sheet	$158 \times ₹66 = ₹10,428$

Registered and eaten are the recorded figures (confirmed); the charge applies posted rates to April (projected, since using the rates this way is an assumption). Four rules sit at the foot of the page: “ate” means a scan against a registration plus any hand-logged scanner failures, shown separately; on a run-out day, eaten is a floor, not a count; the uneaten charge is printed right beside the surplus; and no figure on this page may ever ground a deduction.

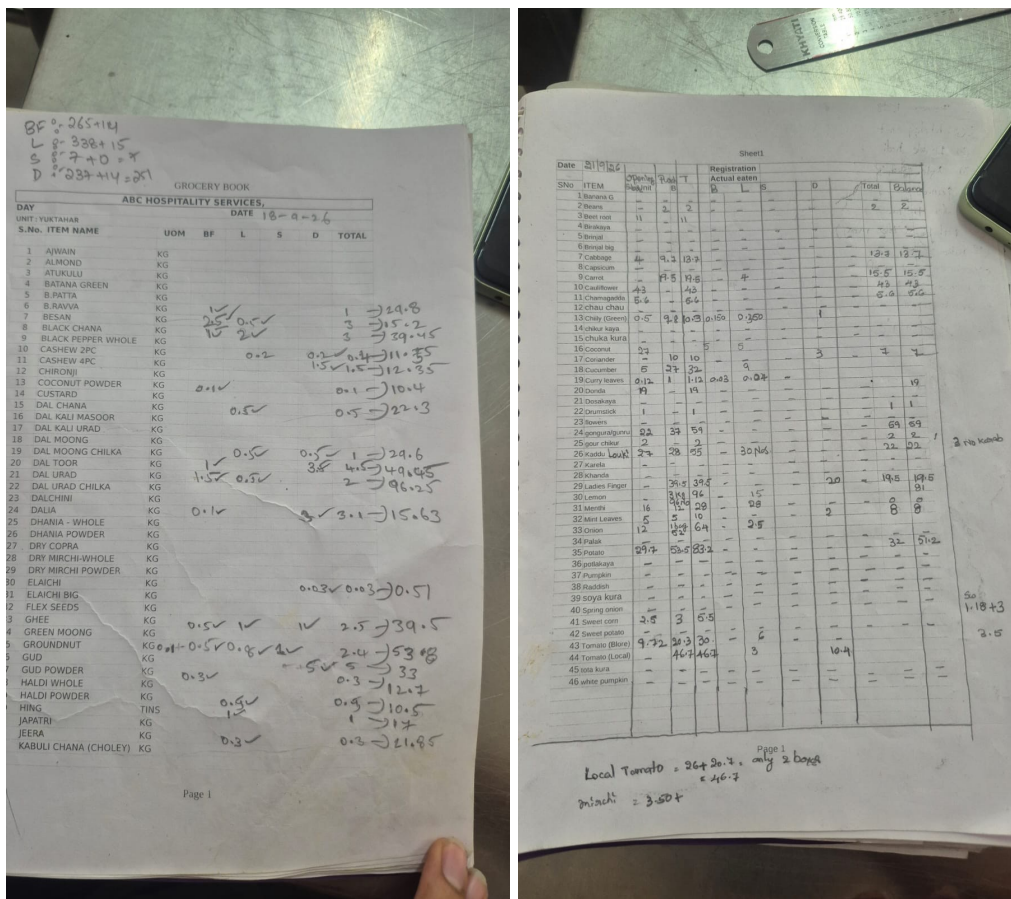
Prototype B: the plan card and daily sheet. The plan card is one line per dish per line, filled in at the evening planning step. Here it’s worked through for the Kadamba veg line on Sunday 19 April 2026, using only the records that would have been available by Saturday evening:

Table 13. Prototype B, the day-before plan card

Step	Rule	Value
Expected eaters	Mean eaten on the same weekday in earlier weeks (5 and 12 April: 168 and 159, confirmed)	164
Share arriving before the late batch lands (9:15)	Earlier Sundays, both lines pooled (confirmed)	54.7 percent
First batch	Expected eaters × share before 9:15 × 1.5 cover	134 portions
Named allowance	About 30 portions for outside diners (single-sourced) and a staff allowance (headcount unknown, assumed)	Added by name, whatever the batch
Float	Extra for bonda, puri, vada and idli (single-sourced)	Per item, on the card
Raw reserve	Kept ready for the late batch and the ordinary top-up	Kept

Portions convert to grams using the existing standard (four idlis a person from 120 grams of batter, 100 grams of sambar; single-sourced). The baseline cooked 70 percent of 817, or 572 portions; with the late batch, the design would have cooked 173 against 179 eaten, leaving just 6 diners to the ordinary top-up (projected). The daily sheet records first batch, late batch, other top-ups, run-out time, what was left over and where it went; at Bakul, a per-item sheet is the whole prototype.

Figure 4. The paper formats Prototype B borrows from Yuktāhār. (a) The Grocery Book for 18 September 2026, where the day-before meeting's output is issued per meal (headed BF 265 plus 14, L 338 plus 15, S 7, D 237 plus 14) with a running balance. (b) The daily vegetable sheet for 21 September 2026, with opening stock, issue by meal and balance. Field photographs, confirmed.



Prototype C: the arrival card and the 9:00 rule. The card is printed straight from the April records.

Table 14. Prototype C, the weekday arrival card, Kadamba, April 2026 (confirmed)

Weekday	In by 8:00	By 9:00	By 9:15	After 9:15	Scans a day, 9:15 to 9:30
Monday	17.7%	50.7%	64.6%	35.4%	117
Tuesday	17.0%	55.2%	68.5%	31.5%	110
Wednesday	22.5%	54.6%	68.7%	31.3%	93
Thursday	21.9%	52.9%	64.5%	35.5%	108
Friday	13.9%	48.2%	62.7%	37.3%	106
Saturday	15.8%	49.4%	62.2%	37.8%	106
Sunday	13.5%	38.9%	54.4%	45.6%	103

The rule: divide diners so far by the day's usual share by 9:00, add 10 percent, subtract the first batch, and start cooking that many portions. On 19 April, 62 veg diners by 9:00 divided by 39.4 percent (the two earlier Sundays) projects about 157, a late batch of about 39. The grace period gets withdrawn if scans after 9:30 stay above April's 9.2 percent (confirmed) for four weeks running.

Prototype D: the Sunday exit on the portal. One control on the existing Skip Meal screen for Sunday breakfast, open until 8 pm on Saturday, offered in one of two forms.

Table 15. Prototype D, the two forms of the Sunday exit

Variant	What the student sees	Charge	Count the kitchen reads on Saturday evening
Opt-in (first draft)	“Confirm Sunday breakfast: you will be charged only if you confirm”	Only on a confirmation	The confirmed count
Evening-before skip	“Skip Sunday breakfast: you will not be charged, and the kitchen will cook less”	Removed on a skip	The registered count minus skips

Whether Skip Meal already lowers the kitchen’s count is an open conflict, so the prototype requires that either form actually does.

Prototypes E and H. The menu note is a half page per operator per cycle, listing the three items most left over, the three that ran out earliest, and items above the gram standard, with residents’ views set beside any proposed pairing. The shared reading itself lasts an hour: each participant writes their own figure first, the page gets read together, and the session records, in the participants’ own words, what would show the belief has actually shifted. Nothing said is attributed outside the room.

Co-design and User Testing (Planned)

These tests are planned for Stage 1, before any pilot begins; none has been run yet. They’re designed to catch a design nobody would actually use, which no model can do on its own.

Table 16. Planned co-design tests (none yet run)

Test (planned)	With whom	What it checks	Result that would change the design
Sunday skip wording	8 to 12 students: skippers, resellers, daily eaters	Understood at once; fear for slot or resale; preferred variant	Misreading: rewrite. Slot fear: guarantee on screen. Preference for confirming: opt-in later
Plan card and 9:00 walk-through	Kadamba supervisor and cook	Card fits the evening step; 9:00 batch without delay; 15-minute items; crest guess	Cannot land by 9:15: 8:45 trigger, higher cover. Too slow: three main veg items only
Ledger review	Yuktāhār’s managers and register keeper	Audit or shared record; fit with paper routine; which “ate”	Audit reading: no per-person entries, monthly threshold. Mismatch: their columns as template
Export effort	CDS staff member	Time for export and page; charge line possible	Over an hour a week: monthly page
Why Bakul residents skip	8 to 12 Bakul residents	Their reasons against the manager’s; pairings; held items	Mostly sleep: menu is no attendance lever. Shared pairings: into the note. Texture complaints: holding time recorded
Bakul reserve walk-through	Bakul manager	Call-to-counter time; basis of the reserve; whether anything is written	Long lead: reserve kept, sized from arrivals. Nothing written: per-item sheet first

The Bakul tests are preparing a later extension, and don't gate the Kadamba pilot in any way.

Scenario Model and Impact Evaluation

The model explores how the system might respond; it doesn't predict. Every output is projected, meaning arithmetic run on real records under stated assumptions. It replays Kadamba's April 2026 breakfast records (confirmed, 32,162 registration rows with scan times), comparing what each rule would have cooked against what was actually eaten, in the real order of arrivals. Kitchen forecasts use only earlier records, so the backtest covers 23 out-of-sample days, 8 to 30 April; student-side scenarios use the whole month. No food cost or waste weight is used; money here is simply the posted rate times counted uneaten registrations.

Table 17. Model assumptions

Input	Central value (tested range)	Basis
Kadamba first-batch ratio	70 percent per line (60 to 80)	Single-sourced; a sensitivity band
Rates	₹48 veg, ₹66 non-veg at Kadamba; ₹53 at Yuktāhār	Confirmed as posted; use for April assumed
Forecast of eaters	Mean of earlier same weekday and line ($\times 0.75$ with an exam note)	Model rule; records confirmed
Arrival profile	Earlier same weekdays (crest 5 lighter to 10 points heavier)	Records confirmed; use candidate
Top-up time	15 minutes (30 in a gas shortage)	Single-sourced; stress assumed
Named allowance; flat projection	About 30 outside diners plus 0 to 30 staff (0 to 60); 25 percent in by 8:00	Single-sourced and assumed
Skip uptake	Charge removed: 25 or 50 percent of no-shows (10 to 75), 10 to 25 percent of skippers still eating (assumed). Nothing returned: zero	Direction confirmed (L2)
Opt-in turnout	60, 72 or 85 percent; 5 percent walk-ins	60 candidate (Yuktāhār June Saturdays); rest assumed
Eaters under a rule change	Held at records (± 10 percent; ± 25 for exams or a favourite item)	Assumed
Adoption and timing	Kitchen weeks 5 to 16; skip proposed week 6, in force week 10	Assumed

Three limits apply throughout. The records cover one hall, one meal and one month, and rest on the operator's own ratio. Kadamba's stated waste of 5 to 10 percent (base unknown) only fits a surplus of several hundred portions if most of it is eaten by others, reheated or never actually cooked, so surplus here means portions cooked beyond what registered students ate, not waste in the strict sense. And demand is censored: on a run-out day, "eaten" is a floor, so an unadjusted forecast would drift downward over time, which is exactly what the ledger's run-out rule is there to answer.

None of this transfers to Bakul, which has no records of any kind. Its manager's account (single-sourced) differs on every kitchen input: a rush at 8:00 to 8:30 with 10 to 15 arrivals in the last ten minutes, cooking about 8 km away from around 3 AM for some items, an unmeasured top-up lead time, and turnout at most about 60 percent, possibly itself an upper

bound (candidate). The arrival card, the 9:00 batch, the 1.5 cover and the Scenario 4 thresholds all belong to Kadamba alone.

Scenario 1. The kitchen rules, backtested. Four rules are replayed across the 23 backtest days.

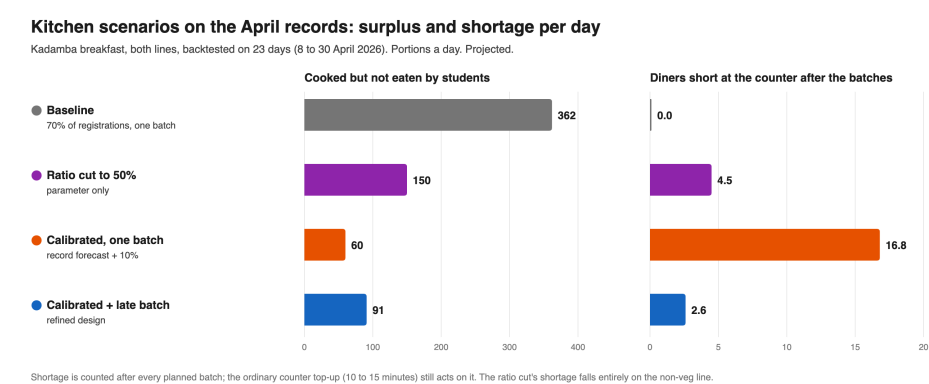


Figure 5. Kitchen rules backtested on 23 April days

Figure 5. Kitchen rules backtested on 23 April days, portions a day, both lines (projected)

Table 18. Kitchen rules on 23 backtest days (projected)

Rule	Cooked a day	Surplus a day	Short a day	Line-days short (of 46)
Baseline: 70 percent, one batch	759	362	0	0
G: ratio cut to 50 percent	542	150	4.5	7
B alone: forecast plus 10 percent, one batch	440	60	16.8	16
B with C: calibrated batch plus 9:00 late batch	485	91	2.6	7

The baseline cooks far ahead of demand (veg runs about 525 a day for 237 eaten). The ratio cut fails on both counts: 150 surplus still remains (138 on veg), and non-veg still runs short on 7 of 23 days, since its turnout reaches 59 percent on busy days. One flat ratio simply can't serve two lines 16 points apart. A calibrated single batch cuts surplus down to 60, but it just moves the cost into the crest instead, 16.8 short a day (13.7 after 9:15), and on 26 April the veg line would have run out at 9:24, 45 short: L14 running in reverse. With the late batch added, 274 fewer portions get cooked, leaving 91 surplus and only 2.6 short, which the ordinary top-up covers; the effective ratio becomes an output ranging from 29 percent (Sunday veg) to 67 percent (Friday non-veg). Across true ratios of 60 to 80 percent, the reduction runs 165 to 382 portions a day: about 105 to 382 net of an allowance of up to 60, and about 214 to 274 at 70 percent.

Scenario 2. Projecting the day from the morning. Three ways of projecting the day's diners from the morning count were tested against what the day actually brought.

Table 19. Scenario 2, error in projecting the day from the morning (projected, on confirmed records)

Projection	Mean error	Mean absolute error	Range of error
Flat, from the count at 8:00	Under-reads by 26.9%	34.8%	–56% to +54%
Weekday profile, from the count at 8:00	Over-reads by 15.6%	28.6%	–35% to +113%
Weekday profile, from the count at 9:00	Over-reads by 2.6%	13.4%	–23% to +70%

A flat projection under-reads because only 13.5 to 22.5 percent of diners have actually arrived by 8:00, not the quarter it assumes. The weekday profile removes most of the bias but not the noise, since the early count is still small, and reading the count at 9:00 instead halves the error again. Hence the 9:00 trigger, with the card rebuilt monthly.

Scenario 3. One Sunday on the veg line. Across three backtest Sundays, the baseline cooked about 563 for about 190 eaten, a surplus of about 373, or 2.5 to 3.4 times what was eaten across ratios of 60 to 80 percent. The refined design would cook about 233, leaving about 45 surplus and about 2 short. Sunday veg is the week’s largest single surplus block, and the natural pilot slice.

One Sunday on the veg line: who arrives, and what each rule cooks

Kadamba, Sunday 19 April 2026: 817 registered, 179 ate (records). Batches are projected under each rule.

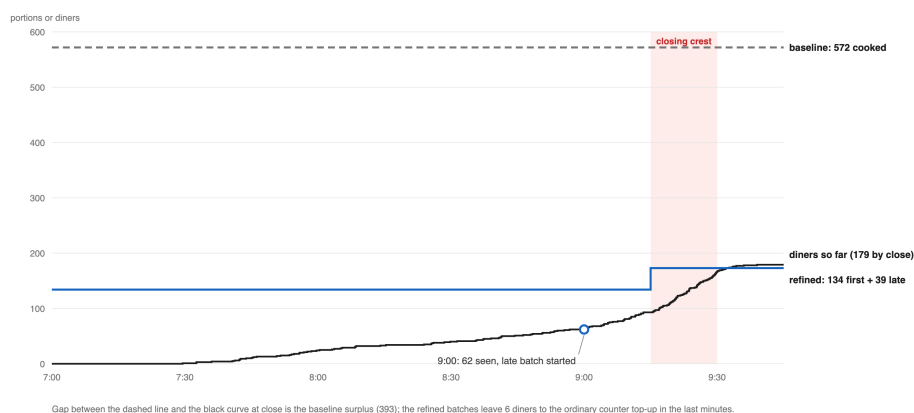


Figure 6. Kadamba veg line, Sunday 19 April 2026

Figure 6. Kadamba veg line, Sunday 19 April 2026: recorded diners against the baseline batch and the refined first and late batches (projected)

Scenario 4. The student side and the default. In April, Kadamba students were charged for 20,221 breakfasts they didn’t eat (confirmed): about ₹7.4 lakh veg and ₹3.2 lakh non-veg, ₹10.6 lakh in all (projected). Kitchen rules don’t touch this number at all.

Table 20. April charges for uneaten breakfasts (projected)

Scenario	Charges	Change
Baseline	₹10.58 lakh	None
Skip returning nothing, any day	₹10.58 lakh	None; unused (L2)
D: Sunday skip, 25 to 50 percent uptake	₹9.78 to ₹10.18 lakh	₹0.40 to ₹0.80 lakh less

Scenario	Charges	Change
Sunday opt-in, turnout 60 to 85 percent	₹9.05 to ₹9.35 lakh	₹1.23 to ₹1.53 lakh less
Every-day skip, 25 to 50 percent uptake	₹5.29 to ₹7.94 lakh	₹2.65 to ₹5.29 lakh less
F (held)	None	₹10.58 lakh off students; bearer per plate basis
Shorter lead, cap of five kept	At most 26.5 percent of no-shows removable	Bound only: 1,072 students × 5 against 20,221

A skip that returns nothing changes nothing, which is exactly why Skip Meal fails today. One that actually removes the charge is the smallest rule change that returns real money, and it scales cleanly from Sunday to every day. The lead-time bound shows that the cap limits any exit, no matter how it's framed.

Table 21. Skip effect a week, by uptake among no-shows (projected)

Uptake	Sunday: returned	Every day: returned	Sunday: fewer registered plates	Every day: fewer registered plates
10 percent	₹4,010	₹24,692	78	472
25 percent	₹10,025	₹61,729	194	1,180
50 percent	₹20,050	₹1.23 lakh	389	2,359
75 percent	₹30,075	₹1.85 lakh	583	3,539

Fewer registered plates only cost the vendor if it's paid on them, which itself is unconfirmed. And if the kitchen keeps its ratio unchanged, opt-in starves the crest: 70 percent of a confirmed count leaves the two lines about 23 short each on Sundays at 72 percent opted-in turnout, and about 65 short at 85 percent, nearly all of it after 9:15, because the ratio silently assumes 30 percent of bookings go uneaten. A skip is less exposed to this than opt-in.

Table 22. Lowest uptake at which a kitchen keeping its ratio runs short (projected)

Ratio	Line	Safe, Sundays	Safe, every day	Days short at 50 percent, every day (of 30)
60 percent	Veg	72.5%	47.2%	1
60 percent	Non-veg	49.0%	3.6%	25
70 percent	Veg	82.3%	66.1%	0
70 percent	Non-veg	67.2%	38.0%	7
80 percent	Veg	89.7%	80.2%	0
80 percent	Non-veg	80.9%	63.8%	0

A Sunday skip is unlikely to cause a crest shortage below about two-thirds uptake on non-veg at 70 percent, or about half at 60 percent, even with a quarter of skippers still eating. An every-day skip at 50 percent uptake runs non-veg short on 7 of 30 days, about 1.4 a day, rising to 3.0 if a tenth of skippers still eat, and 8.9 if a quarter do. So the Sunday skip can go on its own

fixed date, while the every-day skip has to wait until the kitchen actually reads eaters on every line.

Scenario 5. Yuktāhār’s belief against its register. Across nine legible September breakfasts, Yuktāhār registered 228 to 325 and served 41 to 105 (confirmed). Planning on the site head’s “about 50 percent” would cook about 63 portions a day beyond what was eaten, a gap the register itself removes (Sundays 18 to 24 percent, Tuesdays 31 to 40), with uneaten charges of about ₹10,500 a day at ₹53 (both projected). Yuktāhār’s real lever here is a change of belief, not a new routine.

Scenario 6. Sensitivity and stress. Each condition is an assumed shock to Kadamba’s recorded arrivals, unwarned unless stated otherwise. Diners “waiting” are those who arrive after one batch runs out and before the next one lands.

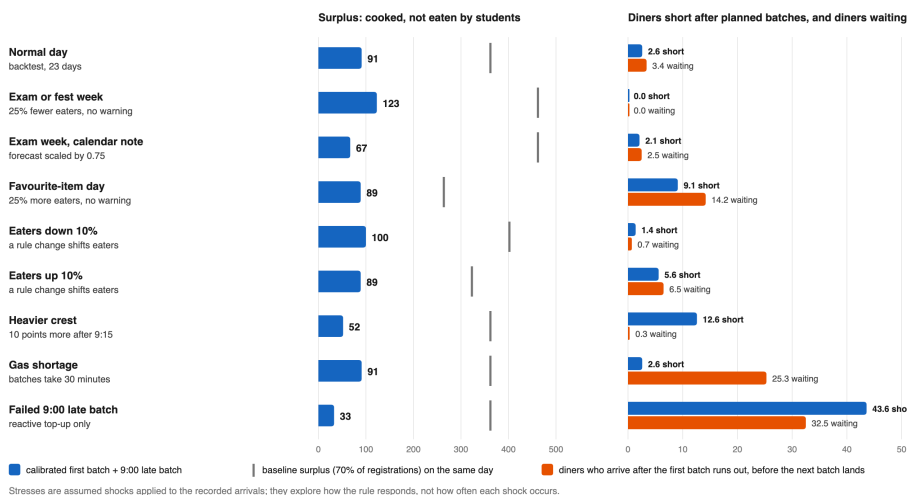
Table 23. Refined design under stress, per day (projected)

Condition	Eaten	Baseline surplus	Refined surplus	Short	Waiting
Normal (backtest)	397	362	91	2.6	3.4
Crest 5 points lighter	397	362	120	0.5	5.7
Crest 5 points heavier	397	362	69	7.7	1.1
Crest 10 points heavier	397	362	52	12.6	0.3
Exam week, 25 percent fewer	297	462	123	0	0
The same, with calendar note	297	462	67	2.1	2.5
Favourite item, 25 percent more	496	264	89	9.1	14.2
Rule change, 10 percent fewer	357	402	100	1.4	0.7
Rule change, 10 percent more	436	323	89	5.6	6.5
Gas shortage, 30-minute batches	397	362	91	2.6	25.3
Failed 9:00 late batch	397	362	33	43.6	32.5

Line-days short (of 46) run 2, 7, 13 and 15 from the lightest to the heaviest crest. The design fails safe toward fewer eaters and loud toward more. The late batch is load-bearing here: if it fails, about 44 diners are short and 33 wait, so the raw reserve is mandatory, and two failed Sundays running should stop the pilot outright. A gas shortage hurts mainly through waiting, and on such days the plan reverts to a larger first batch. A 10 percent shift in eaters keeps surplus between 89 and 100, so the no-adaptation assumption isn’t decisive either way, while the baseline’s surplus is almost pure ratio, running 254 to 471 across 60 to 80 percent.

Stress tests: where the calibrated batch with a 9:00 late batch holds, and where it breaks

Kadamba breakfast, both lines, 23 backtest days. The plan reads normal earlier weeks unless stated. Per day, projected.



Stresses are assumed shocks applied to the recorded arrivals; they explore how the rule responds, not how often each shock occurs.

Figure 7. Refined kitchen design under assumed shocks

Figure 7. Refined kitchen design under assumed shocks on the 23 backtest days, beside the baseline surplus (projected)

Scenario 7. Behaviour over time for the recommended bundle. Adoption pace and timing are assumed; the baseline stays at 362 portions a day (254 to 471) and ₹2.47 lakh a week throughout.

Table 24. The bundle over twenty weeks (projected)

Week	Stage	Surplus a day	Uneaten charges a week
0 to 4	1: routes, tests, joint reading at week 4	362 (254 to 471)	₹2.47 lakh
6	2: Sunday veg pilot; skip proposed	341 (236 to 446)	₹2.47 lakh
8	2	320 (219 to 421)	₹2.47 lakh
10	3: roll-out; Sunday skip in force	262 (182 to 346)	₹2.37 lakh (2.27 to 2.37)
12	3	205 (144 to 271)	₹2.37 lakh (2.27 to 2.37)
14	3; every-day skip decision	148 (107 to 195)	₹2.37 lakh (2.27 to 2.37)
16 to 20	Full adoption	91 (69 to 120)	₹2.37 lakh (2.27 to 2.37); ₹1.23 to ₹1.85 lakh with the every-day skip

How the system might respond over time: the re-sequenced bundle against the baseline

Kadamba breakfast. Weeks from the start of Stage 1. Lines are central values, shaded bands the tested ranges. Adoption pace and rule-owner timing assumed; all values projected.

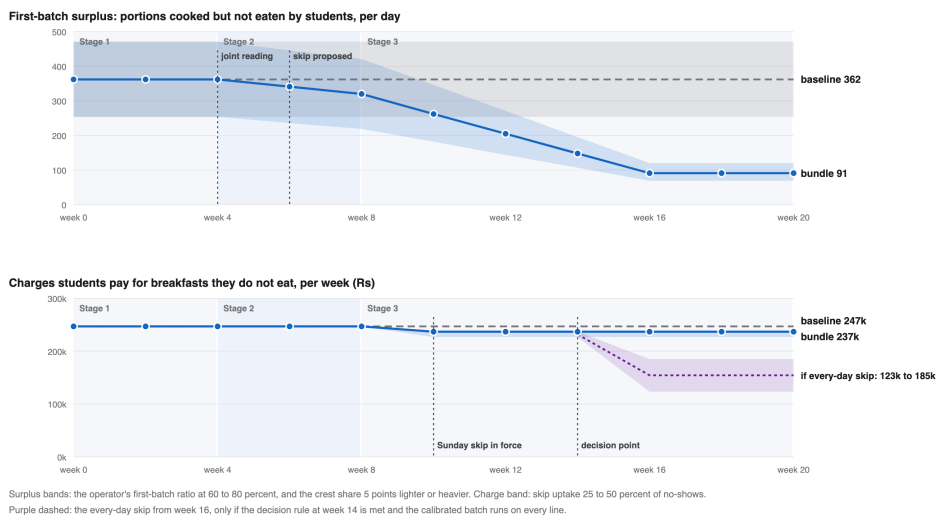


Figure 8. How Kadamba breakfast might respond over time

Figure 8. How Kadamba breakfast might respond over time under the bundle, against the baseline (projected)

Nothing physical actually moves in Stage 1; it’s spent building routes and testing designs. Surplus then falls over roughly three months, perhaps optimistically, since Yuktāhār’s own routine took about four (August to December, single-sourced). The charge only moves when a rule does; a kitchen that improves under a flat charge line is exactly the signature of a quick fix standing in for the fundamental one.

Scenario 8. How each scenario moves every indicator. Figures are projected; directions given without figures are reasoned (assumed).

Table 25. Each scenario across the system

Indicator (today)	Quick fix only	Bundle, Sunday skip	With every-day skip	G	F (held)
Uneaten charges a week (₹2.47 lakh)	Unchanged	₹2.27 to 2.37 lakh	₹1.23 to 1.85 lakh	Unchanged	None; moves per plate basis
First-batch surplus a day (254 to 471)	69 to 120	69 to 120	69 to 120	150	Falls only if kitchen reads eaters
Diners short a day (0)	0.5 to 7.7; 44 if late batch fails	As quick fix	Non-veg short some days unless calibrated	4.5, non-veg	Unchanged
Vendor, if paid on registered plates	Food cost falls	194 to 389 fewer plates a week	1,180 to 2,359 fewer	Food cost falls	Follows served plates

Beyond these headline numbers, every kitchen scenario except G names the staff allowance and reheats less, while supervisor workload rises at first and the share arriving after 9:30 (9.2 percent today) may creep up within its own stop rule. The bundle adds a weekly page and a portal toggle for CDS, and eases pressure on Mess Cell resale; G, by contrast, risks the

allowance and risks trust. Only rule scenarios move the student cost, and only kitchen scenarios move the surplus, so the bundle needs both working together.

2.3 Simulate Stakeholder Responses

Every response below is simulated: a reasoned expectation built from each actor's stated position and incentives, not feedback from the actor itself.

Table 26. Simulated responses to A, B with C, D and H

Stakeholder (grounding)	A: Ledger	B with C	D: Sunday Exit	H: Shared Reading
Near-daily eaters (confirmed)	Indifferent	Fresher food; object to any run-out	Barely affected	Indifferent
Sunday or frequent skippers (confirmed)	Indifferent	Unaffected	Strong uptake if simple and free	Indifferent
Mess Cell resellers (confirmed)	Indifferent	Unaffected	Some shift to skipping	Indifferent
Prism, Kadamba management (single-sourced)	Wary of audit; accepts if blame-free	Receptive as food-cost saving	Opposed if paid on registered plates; neutral if served	Engages if its figures lead
Kadamba supervisor and staff	Fears reprimand	Fears run-outs; accepts with reserve and staff meal named	Unaffected	Curious, if a wrong guess costs nothing
Bakul operator ("there should not be a shortage of food, even if some wastage occurs", single-sourced)	Accepts a check on its judgement; may lack records	Bakul form only; resists any run-out risk	Likely neutral or favourable if paid on plates served (candidate)	Engages once a record exists
Yuktāhār, ABC Hospitality Services	Welcomes if paper-based	Willing host; adopts the 9:00 rule	Welcomes a truer count	Uncomfortable, then anchors on register
CDS (tightened cancellations to cut workload, confirmed)	Accepts a light export	Supportive	Cautious about workload	Receptive if no new work
CFS (knew the figure for years, confirmed)	Receptive to a threshold	Neutral to positive	Supportive if a no-cost pilot	Receptive: its own goal
Student representatives (carried L4's case, confirmed)	Use as evidence	Neutral	Natural carriers	Carry the tested view
Outside diners and staff (single-sourced)	Unaffected	Harmed only if the allowance is unnamed	Unaffected	Unaffected
Facility team at the tasting (observed)	One line if no delay	Unaffected	Unaffected	Neutral

Each response is rational given the actor's incentives, and each incentive can be changed. Prism needs a food-cost saving it actually keeps, visible run-out protection, and a contract

assurance that lower waste alone won't lower its rate. The supervisor needs the blame-free rule and a named staff meal; CDS needs a skip that replaces cancellation work rather than adding to it; and Bakul, whose floor is entirely rational when a late call can't land in time, needs a record of how often a smaller batch would actually have run short. The simulation makes the case for one joint reading rather than separate proposals, and against delaying the booking change behind the kitchen. If Bakul is paid on plates served, its likeliest objection is shortage risk, not a lower booking count (simulated, candidate).

2.4 Test Interventions Against System Objectives (Deliverable 4: Evaluation Framework)

The objectives come from the reframe, and belong to the system rather than any one stakeholder: notice its own cost, cut the first batch without trading waste for run-outs, make no one worse off, and be usable. Failing any of four gates makes a concept ineligible, whatever its score.

G1, absorbers kept: resale, reuse, outside-diner and staff meals, walk-ins, late service and outside venues all survive, and only the over-cooked batch, reheating and the bin are targeted (P5). **G2, safety kept:** every food-safety step stays intact (P4). **G3, aimed at structure:** it neither asks students to behave better nor blames operators, and it isn't gated on an unknown that could reverse it (P3; LP10 held). **G4, quick fix paired:** any kitchen lever comes with a route, default, goal or belief lever landing on a date independent of kitchen performance.

Table 27. Criteria, scored 0 to 4 and weighted

Criterion	Result measure	Weight
O1: Booked against eaten	Uneaten charges; share of registrations uneaten	15
O2: Cooked against eaten (only with O3)	First-batch surplus a day	15
O3: Crest service, every hall	Shortage after all batches per hall and line; a residue under about ten a day counts as protected	15
O4: Fundamental capability	Route built; L11 at Kadamba; goal indicators adopted	15
O5: Fairness	Who bears any new cost	10
O6: Food quality	Share from a fresh or late batch; holding time; grams per plate; taste and texture	10
O9: Desirability	Simulated response; co-design tests once run	10
O7: Feasibility now	Who must act	5
O8: Reversibility and learning	Scope of a first pilot	5

The weights are the team's own judgement. Food quality rose from 5 to 10, because reheated surplus is a cost students actually eat, and the quickest way to cut surplus is worse portions; desirability entered at 10; feasibility and reversibility fell from 10 to 5, and cooked-against-eaten from 20 to 15, because weighting too heavily toward what a student team can do pulls every score toward the quick fix. Process checks are monitored, never scored. O2 is never read on its own, and O6 now counts holding time and texture too, after the Bakul manager reported that aloo paratha carried over from the Hafizpet kitchen loses its texture (single-

sourced). No score moved because of this, since everything scored here still rests on Kadamba's records. Scores run from 0 (worse) through 1 (no effect) to 4 (large gain shown in the model); on O3, a 3 means a residue that one top-up can cover, and at Bakul any concept is judged against its held reserve. The total is out of 400, shown as a percentage.

Table 28. Scoring against the criteria (0 to 4 per criterion; total out of 400 as a percentage)

Concept	Gates	O1	O2	O3	O4	O5	O6	O9	O7	O8	Score	First draft
A: Breakfast Ledger	Pass	2	2	4	4	2	1	2	4	4	68%	74%
H: Shared Reading	Pass	2	2	4	3	3	1	2	3	4	65%	New
F: Billing at consumption	Fails G3 (held)	4	2	4	2	3	1	2	0	2	63%	61%
C: Crest Plan, alone	Fails G4 alone	1	2	4	1	3	3	3	4	4	63%	64%
D: Sunday skip, ratio kept	Pass	2	2	4	2	3	1	3	1	3	60%	59%
E: Operator Menu Slot	Pass	1	2	4	2	2	2	3	3	4	60%	61%
B: Kitchen Kit, alone	Fails G4 alone	1	4	1	3	2	3	1	3	4	58%	65%
D: every-day skip	Pass	3	2	1	2	3	1	3	0	2	50%	New
D: Sunday opt-in (first draft)	Pass	3	3	0	2	2	2	2	1	3	50%	51%
G: Ratio cut to 50 percent	Fails G1, P4	1	3	3	1	1	1	1	4	4	48%	58%
First-draft bundle (skip waits for the kitchen)	Fails G4	2	4	3	4	3	3	3	3	3	79%	80%
Re-sequenced bundle (A, H, Sunday skip on fixed dates; B with C as quick fix)	Pass	2	4	3	4	3	3	3	3	3	79%	New

A leads because it builds the route every other lever needs, even though it moves no food or money on its own. H acts deepest of all, but the discomfort of being shown one's own figure caps its O9. B alone has the largest physical effect (surplus down from 362 to 60) and the worst crest performance (16.8 short), which drags it from second place down to seventh. The every-day and opt-in forms of D move ₹2.65 to ₹5.29 lakh and ₹1.23 to ₹1.53 lakh a month respectively, but both fail the crest test (O3 at 1 and 0). G fails the gates outright and still leaves 150 surplus portions on the table. The two bundles hold identical parts and score identically, because a weighted score can't see sequencing, but the gate can. The first-draft bundle made the Sunday skip wait for the kitchen to prove itself first, which is the shifting-the-burden pattern playing out in miniature; Scenario 4 shows that wait is unnecessary, so the re-sequenced bundle passes G4 and is the one recommended.

2.5 Identify Unintended Consequences

Several second-order effects here are really just the loops themselves pushing back.

Table 29. Unintended consequences and mitigations

Intervention	Consequence	Loop	Mitigation
B, C	Food taken from outside diners and staff	L9 (candidate)	Named allowance, recorded as served
B	Crest run-outs on heavy days	L14 reversed, L12	Late batch, 1.5 cover, reserve, menu days noted
B, C	One loud shortage restores the hedge	L14 ratchet	Judge over months; every run-out recorded
B, A	Better kitchen hides the gap upstream	L17 (candidate)	Ledger carries the charge; kitchen labelled quick fix
C	Late batch lands in the busiest quarter-hour	L12	Live or steamed quick items; walk-through first
B	Exams and events break the forecast	L7	Calendar note
D	Resale buyers lose Sunday slots	L1	Accepted; no market closed
D	Opt-in with a fixed ratio under-cooks the crest	L13, L14	Skip keeping the default
B, C	Less reheating changes what students get	L13 (open)	Recorded on the sheet
B, C at Bakul	Kadamba rule trims a batch no top-up can replace	L12 with delay	No Kadamba rule applied; floor kept
C at Bakul	Bigger reserve softens held items (candidate)	L13 holding (open)	Reserve sized from own arrivals; texture under O6
A	CDS workload rises	Governance	Paper-first; tested time budget
A, D	Two people on record as head of CDS	L10	Send by role

Table 30. Policy resistance (likelihood assumed)

Adaptation	Loop	Likelihood	Early warning	Mitigation
Vendor raises its rate at renegotiation	L8	Medium on registered plates, low on served	Rate proposals	Ask plate basis first; show food-cost saving beside lost plates
Skip, then eat on another's scan	L1	Medium Sundays; higher daily	Eaten per registration rising	Walk-in ₹81 above ₹48 (confirmed); daily step behind its rule
Smaller portions, hidden leftovers, reheated stock	L11 goal; L13	Medium	Grams per plate; "reuse" rising	Never read alone (P7); store issues reconciled
"Ate" gamed via hand-logged scans	L10	Low to medium	Hand-logged share rising	Own line; a rise triggers a scanner check, not a staff check
Allowance quietly cut	L9	Medium, least visible	Allowance below plan	Missed meal is a stop condition

Adaptation	Loop	Likelihood	Early warning	Mitigation
CDS re-tightens cancellations	Governance (December 2024 precedent)	Low to medium	Proposals to cap skips	Skip replaces cancellation work; CDS in from week 4
Censoring spiral lowers forecasts	Candidate chain on L11	Medium	Repeat weekday run-outs	Demand read as count at run-out ÷ card share; two run-outs raise the forecast
Ledger read as an audit	Accidental adversaries	Medium at Kadamba	Round numbers; “left over” always small	Blame-free rule; review, never deduction
Hedge quietly restored	L14 ratchet	Medium to high	Effective ratio drifting to 70 percent	Weekly ratio line
Grace period softens the close	L15 (candidate)	Medium	Share after 9:30 above 9.2 percent	Withdraw grace after four weeks; late batch stays

2.6 Refine the Intervention

Tests, stress runs, consequences and the Bakul evidence together changed the first concepts in twelve ways.

Table 31. Refinements

First concept	What changed it	Refined version
B alone	16.8 short a day; 45 short at 9:24 one Sunday	B with C: batch for diners before 9:15, late batch at 9:00
C projecting at 8:00	Error 35 percent flat, 29 profile at 8:00, 13 at 9:00	Projection at 9:00
10 percent first-batch buffer	About 18 a day wait	1.5 cover: 3 wait, 2.6 short, surplus 91
One C for every hall	Bakul’s rush, phoned top-up, held reserve	C per hall; Bakul reserve once records exist
Kitchen bundle as the design	Shifting the burden	Labelled the quick fix everywhere
Sunday skip after the kitchen reads eaters	Scenario 4: safe below two-thirds uptake	Fixed date, week 6; every-day skip on a rule
Sunday opt-in	23 to 65 short; ₹81 for the forgetful	Evening-before skip; opt-in later on the path
Any skip	Returning nothing, it is unused (L2)	Must remove the charge and lower the count
A without the charge	Surplus read as solved	Weekly charge line
B without allowance	G1	Named allowance on every card
A as dashboard	Yuktāhār dropped a paid ERP (single-sourced)	Paper-first A4 ledger
Records alone	Records cannot show use	Planned co-design tests

Table 32. Late-batch design grid (projected)

Start	Cover	Buffer	Surplus	Short	Waiting
8:45	1.1	10%	77	10.6	20.0
8:45	1.5	20%	116	4.9	6.2
9:00	1.1	10%	67	8.6	17.7
9:00	1.3	10%	75	6.5	7.4
9:00	1.5	10%	91	2.6	3.4
9:00	1.5	20%	118	0.8	3.4

The choice made is the least surplus that keeps waiting under five a day. Capping the day's total at 1.5 times the forecast would save 21 more portions for 2.7 more short (projected), left as an option for the pilot to decide.

2.7 Deliverable 5: Final System Design Recommendation

The Recommended Intervention

Route and read the gap together, open the Sunday exit on a fixed date, and calibrate the kitchen to buy time. The kitchen bundle runs alongside all of this as the quick fix, and is named as such wherever it's reported.

Table 33. The recommended sequence

Stage	Weeks	Fundamental fix, on fixed dates	Quick fix	Leverage points (places)
1: Routes and joint reading	0 to 4	Ledger with charge line; co-design tests; week 4 joint reading; goal and indicators to CFS and CDS	Cards and sheet drafted with the Kadamba supervisor	LP1 (6), LP11 (2), LP12 (3)
2: Exit proposed; Sunday veg pilot	5 to 8	Week 6: Sunday skip to the rule owner via student representatives, with the plate-basis question; shared readings	Kadamba Sunday veg runs card, allowance, 9:00 batch and sheet; Yuktāhār re-anchors; tasting line if agreed	LP2 (5), LP11; LP5 (6, 8), LP6 (6, 10), LP3 (3, 8), LP7 (8)
3: Skip in force; roll-out	9 to 20	Skip from week 10 if adopted; week 14 decision; handover from week 16; exit at week 20	Every day, both lines; forum on a cycle; menu notes	LP2, LP4 (4), LP8 (6); LP5, LP6
Held	Until facts known	Billing and payment basis; outcome owner	None	LP10 (5), LP9 (5)

Bakul is a later hall, sitting outside this pilot: a third regime with no records, and one the Kadamba model simply doesn't transfer to. The recommendation extends there only once Bakul's own scan export gives weekday turnout and arrivals, the call-to-counter lead time has been measured, an item-level record sits behind the weekly review, any Bakul kitchen lever reduces waste without raising shortage risk, and, for the Sunday exit specifically, its plate basis (reported as plates served, single-sourced) is confirmed.

The Architecture Path

The path moves from a default that has to be escaped, to a booking that actually matches intent, one reversible step at a time.

Table 34. Decision points and feedback rules

Decision point	When	Step	Rule	If evidence goes the other way
DP1: Sunday skip	Week 6, fixed	Proposal; in force about week 10	None; goes on this date	If declined, the Sunday charge line goes to the escalation forum
DP2: Every-day skip	Week 14	From about week 16	Sunday uptake at least 25 percent; no Sunday over 10 short; eaten per remaining registration rising only a few points; calibrated batch on both lines	Uptake under 10 percent: retest wording. Shortage: that line reads eaters first
DP3: Default matched to intent	After a term of DP2	Opt-in on lowest-turnout days; no automatic registration off campus	Skip used by over half of no-shows; kitchen reads eaters everywhere; never costs a hall slot	Low uptake, stable charge: keep skip and default
DP4: Billing and payment	Plate basis known	Billing at consumption, or payment on served plates	Basis known operator by operator; resale reliance checked	If registered plates and no rate agreement: billing stays; the skip carries students

Theory of Change

Table 35. From root cause to expected response

Root cause	Bundle change	Expected response
Gap measured but unrouted (L10; L4 fires on loudness)	Ledger, charge line, threshold	Cost becomes a trigger beside complaint (candidate)
Registration count sought while minimising wastage is stated	Goal carried with two indicators	Indicators cited in a decision (untested)
Beliefs holding count, ratio and default	Shared readings	Anchors move toward records (candidate)
Fixed ratio, no learning record at Kadamba (L13, L11 absent)	Plan card and sheet create L11	Surplus 254 to 471 down to 69 to 120 a day (projected)
Top-ups blind to the crest (L12, L14)	9:00 late batch	0.5 to 7.7 short; 9 to 13 on heavy days (projected)
Count as a paid obligation (L8, L17)	Forum and food-cost case	Kadamba's own reason to cut (candidate)
Default with a narrow exit (L13, L2, L1)	Sunday skip, then every-day on its own rule	₹4,000 to ₹30,000 a week on Sundays; ₹62,000 to ₹1.23 lakh a week every day (projected)
Billing at the booking (L1, L6, L8)	Held; ledger builds the case	Decided once the basis is known

The bundle works on the place-3 goal, moving the working measure from the registration count to cooked-against-eaten. That's the very aim the institution's own poster already names.

Interventions Across the Iceberg

At the level of events, the daily sheet records batches, run-outs and leftovers, and marks run-out days as censored, flagged by run-out time per line. At the level of patterns, the monthly arrival card and the ledger's behaviour-over-time lines are watched through weekly surplus, shortage and charge, the share after 9:30, and the effective ratio. At the level of structures, the ledger route, plan card, skip and stated goal are tracked through whether the ledger actually reaches the rule owner, skip uptake, and the two indicators. And at the level of mental models, "registration is demand," "breakfast is about 50 percent," "cooking to registrations is what we are paid for," "we cannot nullify it," "they are catching us," shared readings address each holder directly, with Bakul's beliefs held as open questions, and the indicators are shift signals such as the site head's figure finally matching his own register.

Pilot Design and Monitoring

Table 36. Pilot design

Element	Specification
Where and when	Kadamba veg line, Sunday breakfast: four Sundays in Stage 2, continued in Stage 3, judged at week 16; Yuktāhār for re-anchoring
Convening	Joint reading at week 4 agrees the definition of "ate" and hears the Sunday proposal; repeated at week 12
Runs and agrees	Kadamba supervisor runs it; the team fills the ledger. Prism's operations manager, ABC Hospitality Services, the facility team's parent body and the rule owner agree
Result measures	Uneaten charges; surplus; run-outs and diners short; grams, taste and texture; holding time; fresh or late share; allowance meals
Process checks	Sheet nine days in ten; ledger weekly; late batch by 9:05; sessions on date (never reported as results)
Comparison	Sunday non-veg, Saturday veg and earlier Sundays
Success	Sunday veg surplus from about 370 to under 100 (model about 45); no Sunday over 10 short; run-outs no more frequent than comparison days; every allowance meal served; grams and taste held; safety recorded; no sheet-based reprimand; charge line reported weekly
Stop	Shortage beyond one top-up two Sundays running; late batch failing twice; any allowance meal missed; grams below standard; any figure used against staff; consent withdrawn
Would change the plan	Base of the waste figure; whether Skip Meal lowers the count; plate basis; co-design results

The team must be careful not to become the intervenor the system quietly comes to depend on. The CDS office compiles the page by role, operators' own data roles fill their sheets, and the escalation forum receives the threshold; a named outcome owner (LP9) takes over once the head of CDS is actually known. CDS compiles alongside the team from week 16 and alone from week 18; the team steps back at week 20, once two pages and one forum have run without it. Each archetype is checked monthly for at least six months after that.

Table 37. Monitoring after handover

Archetype	Indicator	Signal of return
Shifting the burden: reliance on quick fix	Kitchen-led fall in surplus against fall in charges	Surplus falling, charge flat for three months
Shifting the burden: fundamental progress	Decision points on date; indicators cited; uptake	A decision point missed twice
Fixes that fail	Effective ratio, weekly per line	Drift toward 70 percent for four weeks
Soft close (L15)	Share after 9:30	Above 9.2 percent for four weeks
Accidental adversaries	Completion; round numbers; hand-logged share	Completion under nine in ten for two weeks

Every waste indicator is read alongside the run-out indicator for the same hall and line. Some results look better before they look worse (a bad crest can bring the hedge straight back months later), and others look worse before better (supervisor workload in Stage 2, the charge staying flat until week 10), so each expected shape is written down on the ledger in advance.

Who Must Act

Of thirteen actions in total, the student team can complete two entirely on its own: building the ledger, cards and sheet, and running the co-design tests. It can convene or support four more that others have to decide: the joint and shared readings (CDS as convenor), taking Yuktāhār's routine to Kadamba, the calibrated and 9:00 batches (Prism's supervisor decides), and Yuktāhār's re-anchoring. It can carry six forward as evidenced proposals: the tasting line (facility team, parent body unknown), a cost threshold (Student Council, Parliament, CDS Committee), the stated goal (CFS and CDS), the Sunday exit (rule owner, Mess Office or CDS, unconfirmed), the billing or payment basis (only once the plate-basis question is answered), and an owner of booked, cooked and eaten (CFS and CDS). And it must complete one thing outright: the handover, by week 20.

Before and After

Read as a range, first-batch surplus falls from about 254 to 471 portions a day down to about 69 to 120, over roughly four months, and the weekly uneaten charge falls from about ₹2.47 lakh to ₹2.27 to ₹2.37 lakh with the Sunday skip in place, and to ₹1.23 to ₹1.85 lakh if the every-day skip follows (all projected). The kitchen change is large and the student-side change stays small until the rule actually moves, which is really just the diagnosis restated as a system response. Kitchen levers reduce surplus and run-outs; only a rule moves the booking itself; and the design keeps both in view at once, so the first is never mistaken for the second.

3 Key Findings

1. The system is missing routes, not measurement. The enabling lever sits at place 6 of 12: routing records that already exist to the kitchen and to the rule owners, and a student team can prototype it right now as a paper ledger.
2. The loops form a shifting-the-burden structure. The kitchen levers are the quick fix, and the booking, goal, belief and route levers are the fundamental fix, so the

recommendation runs the fundamental fix on fixed dates, whatever the kitchen pilot shows.

3. The deepest levers are the beliefs that let the registration count stand in for demand (place 2) and the working goal the system actually seeks (place 3). The deepest rule lever that isn't gated on an unknown is the breakfast default and its exit (place 5), provided no skip ever costs a student a hall slot.
4. Cutting Kadamba's ratio to 50 percent fails on both counts, leaving about 150 surplus portions a day and starting non-veg run-outs. A calibrated first batch on its own just moves the cost into the crest; add a 9:00 late batch, and first-batch surplus falls from about 254 to 471 portions a day to about 69 to 120, with a residue the ordinary top-up covers (all projected). The design fails loud when eaters exceed the forecast, so the raw reserve and the stop rules are essential parts of it, not add-ons.
5. Kitchen levers leave the student cost completely untouched: ₹10.6 lakh was charged in April alone for uneaten Kadamba breakfasts (projected). Only a rule that actually returns something moves that number. A Sunday skip is unlikely to run the crest short below about two-thirds uptake on non-veg, so it can go on a fixed date now, while the every-day skip waits on its own decision rule (projected).
6. Kadamba and Yuktāhār show that a learning practice is possible under the very same rules, but the two halls differ in five other ways and still leave a similar breakfast gap, so the cause of that difference stays a candidate reading, not a conclusion. Targets are anchored outside each hall's own past and paired with portion, quality and run-out measures, and no lever may cut waste by raising shortage risk.
7. Every modelled figure in this report is Kadamba's. Bakul, the third regime, has no records at all. It needs its own scans, a measured top-up lead time and an item-level record before any lever can be sized there. Sunday breakfast on Kadamba's veg line is the first test slice.